Python 3 Fundamentals & Scripting Developer Quick Reference

Category: Python Programming • Lead Mentor: Nasir • Licensed for Educational Study

1. Core Data Structures & Idiomatic Python

- Lists, Tuples, and Dictionaries: Choosing mutable vs immutable collections based on performance.
- List Comprehensions: [x**2 for x in numbers if x % 2 == 0] for readable, efficient transformations.
- Context Managers: with open("data.json", "r") as f: Ensures file handles close automatically even on exceptions.

2. Working with REST APIs & Web Scraping

- The requests library: Handling GET/POST requests, HTTP headers, authentication tokens, and timeouts.
- JSON handling: json.loads() and json.dumps() with indent formatting for structured API payloads.
- Virtual Environments: Always isolate project dependencies using python -m venv venv.